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Glossary

See the Glossary for a list of commonly used terms and definitions.

Overview

Getting Started

Downloading and Installing

OS/Platform Differences

- Keyboard shortcuts

UI Overview

Scaling

“the whole UI scales with the window width, against a 3840 pixel reference. A window 3840 wide renders at scale 1.0, and a window 1280 wide renders everything, fonts included, at one third of that. Nothing is tied to the display DPI. If you work in a small window, raise UI Scale and Font Scale in Settings to compensate; they multiply on top of the automatic factor.”

Your First Project

Understanding File Types

Supporting Models

Supporting Overview

Basic Workflow

CONTENT NEEDED

Features

Support Basics

CONTENT NEEDED

- Horizontal line of layers

Manual Supports

CONTENT NEEDED

Left Menu (REWORD)

The left icon panel controls many support options.



Figure 1: Left panel icons

Symmetry

CONTENT NEEDED

Path Supports

Path supports are created on a line between two or more supports. A path is drawn from the tip of each support in the order they were selected. (Use F3 + LMB when selecting multiple supports.)

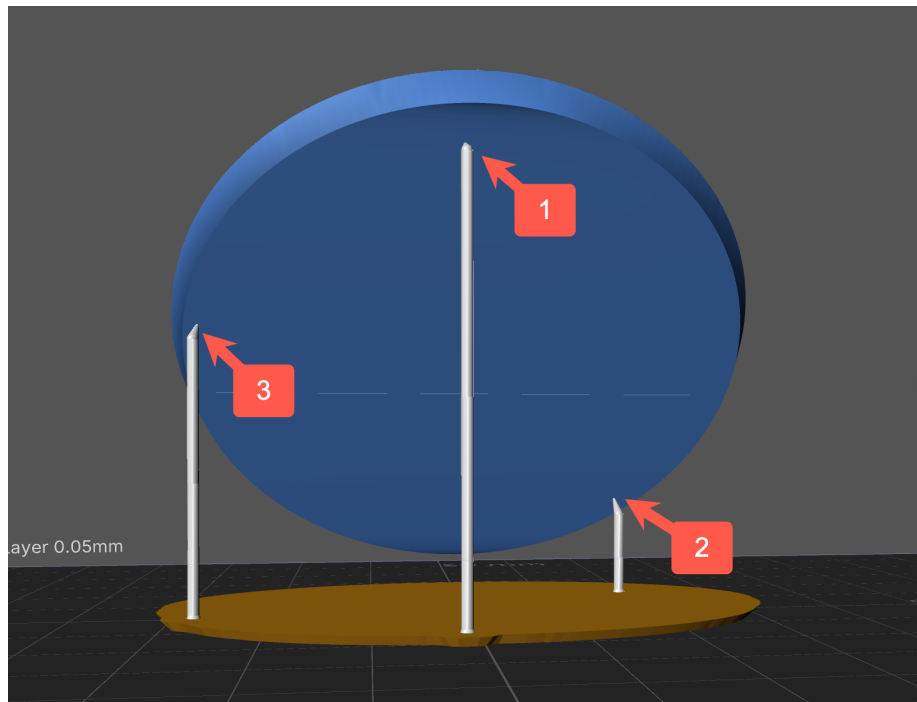


Figure 2: Round Base with Three Supports

Settings controlling spacing of supports for the following features are located in the right menu under Advanced Placement => Line and Fill.

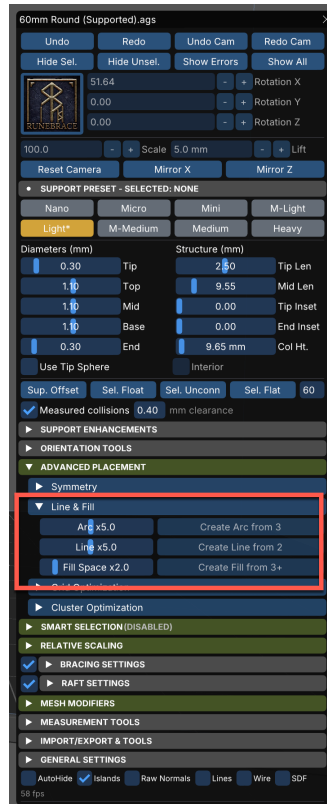


Figure 3: Right menu with line and fill highlighted

Important Considerations

Settings in this section are not distances in millimeters; they are multipliers of the support diameter. Runebrace determines how many supports fit in the space between the selected supports, then spreads generated supports evenly along the segment.

Very small values stop having an effect at some point, and are ignored.

A segment too short to fit anything gets no supports.

Line

Adds support along a line between two and only two supports.

Line placement is available through the right menu and the left button menu (fig. 1).

Spacing for supports along the line is controlled by the right menu under Advanced Placement => Line & Fill => Line.

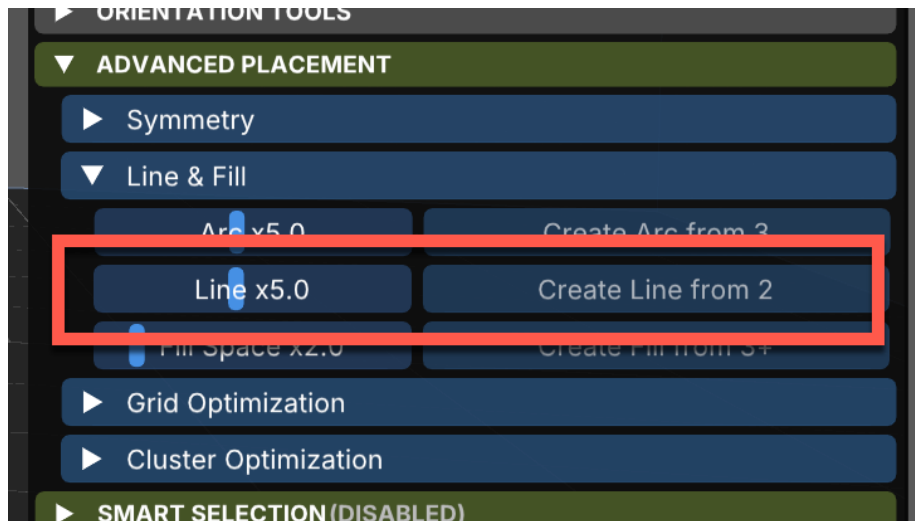


Figure 4: Menu for adjusting Polygon Fill spacing

Selecting support #1, then #2, then creating Line supports results in this:

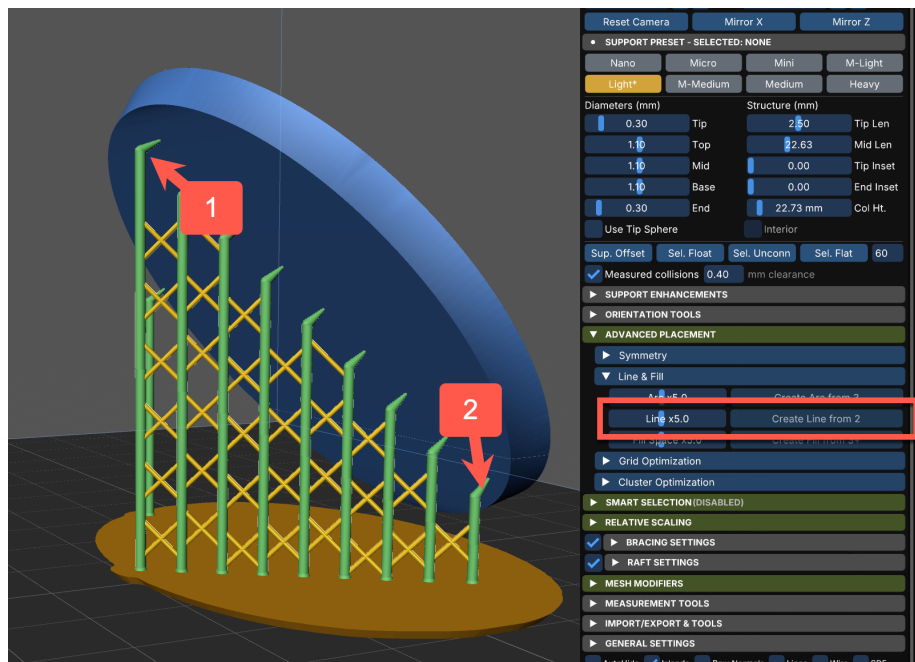


Figure 5: Supports created via line

Arc

Using Arc adds supports along three selected supports—no more, no fewer.

Arc placement is available through the right menu and the left button menu (fig. 1).

Configure using the settings in the right menu Advanced Placement => Line & Fill => Arc.

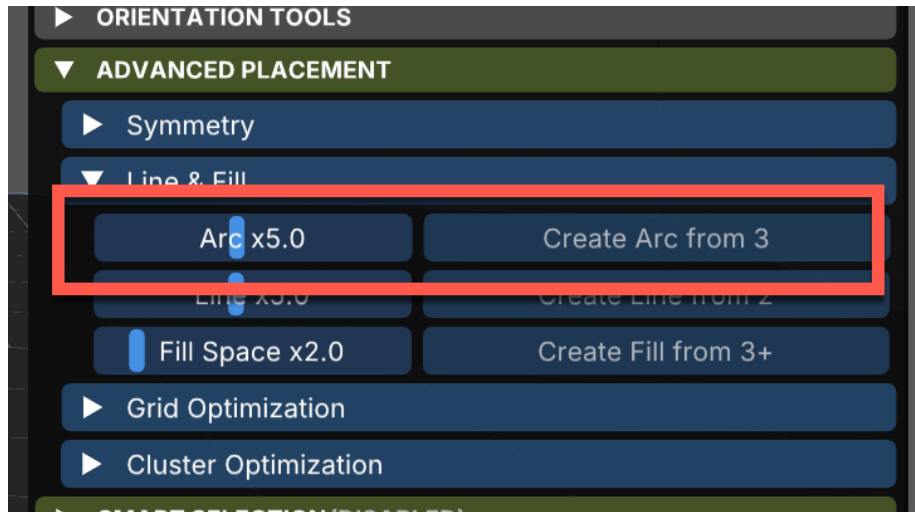


Figure 6: Settings for Arcs

Selecting support #1, then #2, then #3, then creating Arc supports results in this:

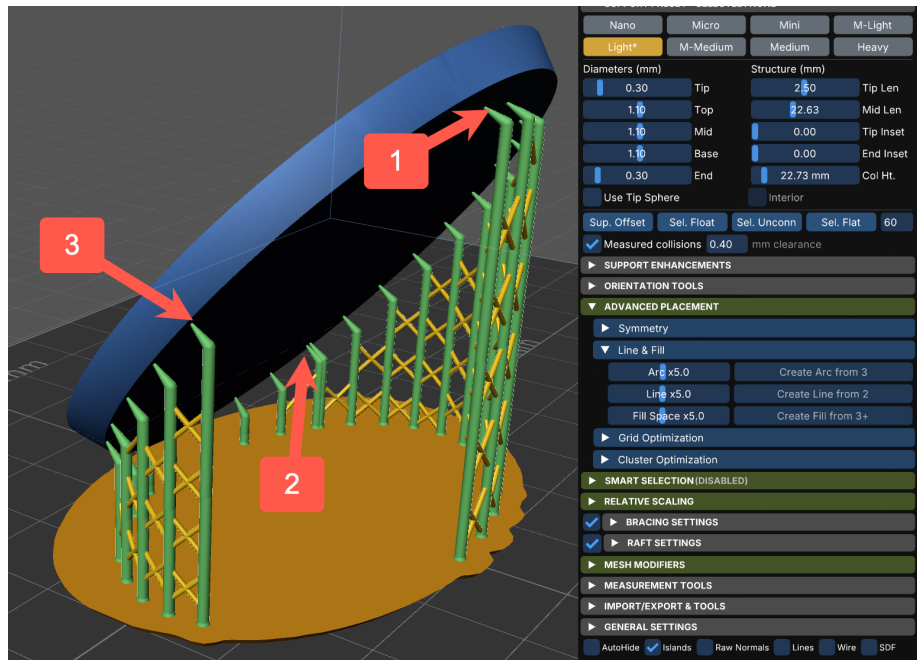


Figure 7: Supports created via Arc

Note the double supports appearing at callout #2. This is due to the spacing along the arc, with one added support being in close proximity to the original #2.

Circle

Using Circle adds supports along a circle path drawn through three supports—no more, no fewer.

Circle placement is available **only** through the left button menu (fig. 1).

Configure using the settings in the right menu Advanced Placement => Line & Fill => Arc as shown above in fig. 6.

Selecting support #1, then #2, then #3, then creating Circle supports results in this:

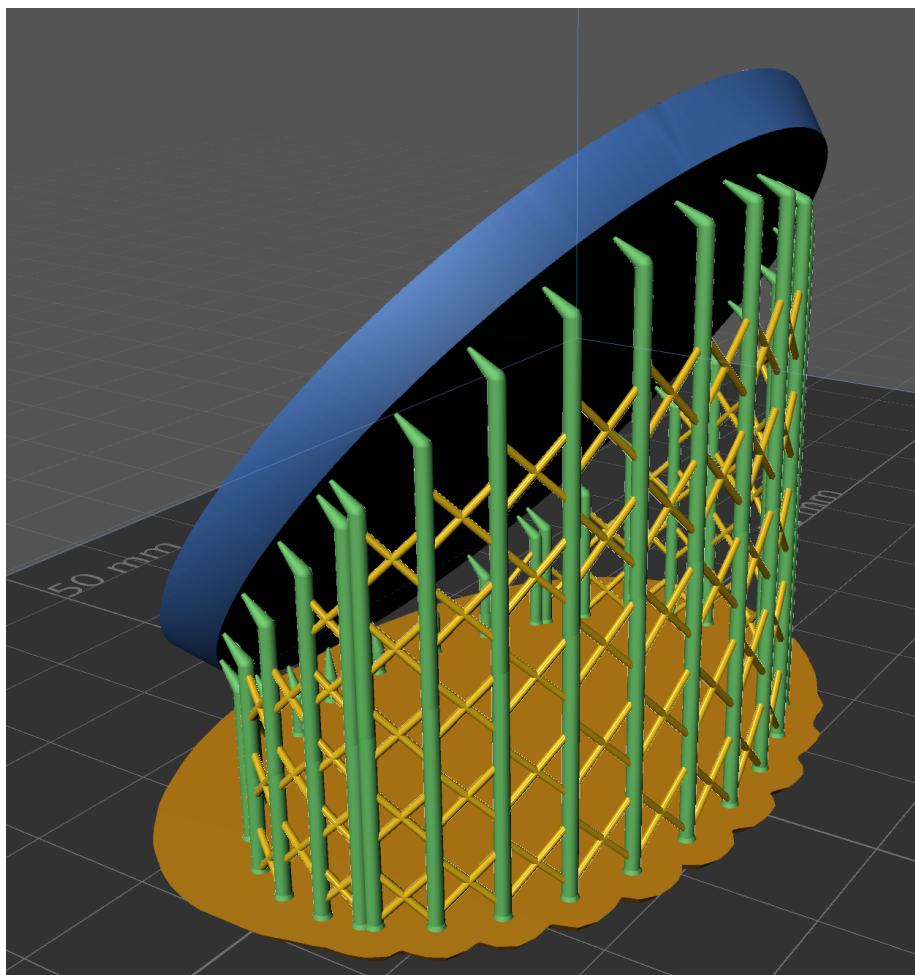


Figure 8: Supports created via Circle

Again, note the doubled supports, this time at #2 and #3.

Fill Basics

Runebrace also provides area fill capabilities.

We'll start with a basic example, then show a more complex, practical example after. Triangle Painting and fills are covered separately below.

Each feature below will show results starting from the above figure fig. 2, a round base with three supports selected in the order they're labeled.

Polygon Fill

Polygon Fill, commonly referred to as Poly Fill, fills an area defined by three or more supports. No rim support is created.

Poly Fill is available through the Fill Space option in the right menu and the corresponding button on the left button menu (fig. 1).

You can also use the Fill Space button from the right menu (see below).

Using Poly Fill/Fill Space from our starting point results in this:

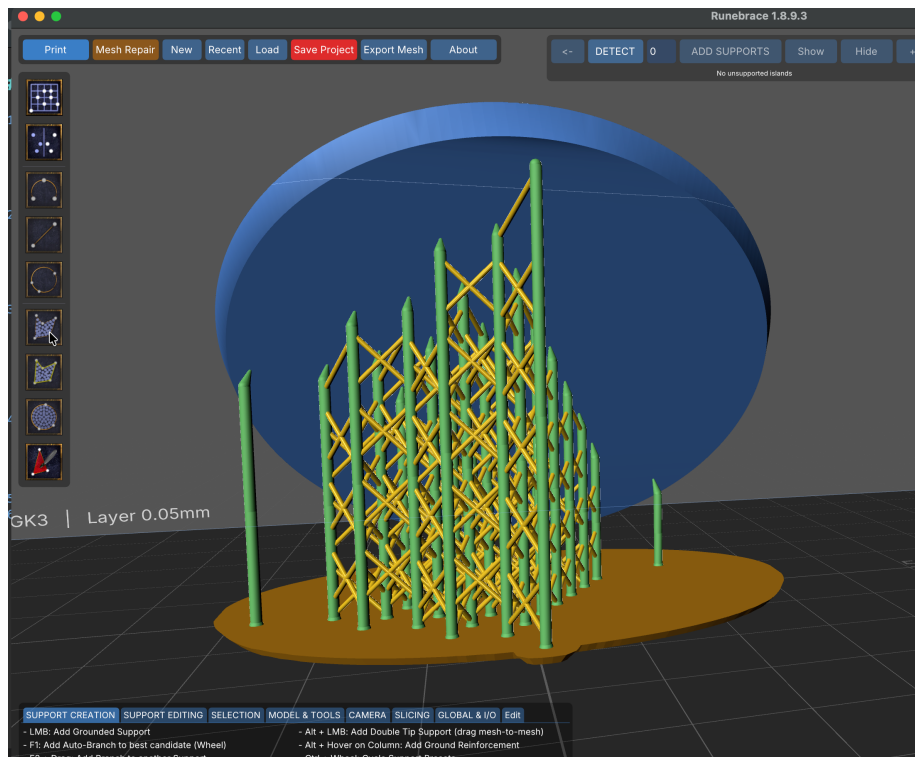


Figure 9: Poly fill

Why: EDIT: ADD USE CASE

Adjust the spacing of the filled supports under Right Menu => Advanced Placement => Line and Fill => Fill Space

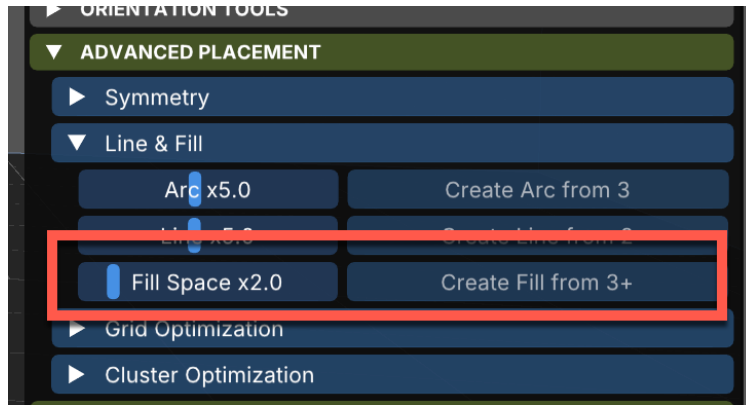


Figure 10: Menu for adjusting Polygon Fill spacing

Perimeter and Fill

Perimeter & Fill does the same as Poly Fill but also lays supports along the area's bordering path itself. Note the menu button's icon shows this via the yellow outline.

Poly Fill is available **only** through the corresponding button on the left button menu (fig. 1).

Select three or more supports, then press the Perimeter and Fill button (there's no alternative from the right menu). One action creates the perimeter and inside area supports.

Using Perimeter and Fill from our starting point results in this:

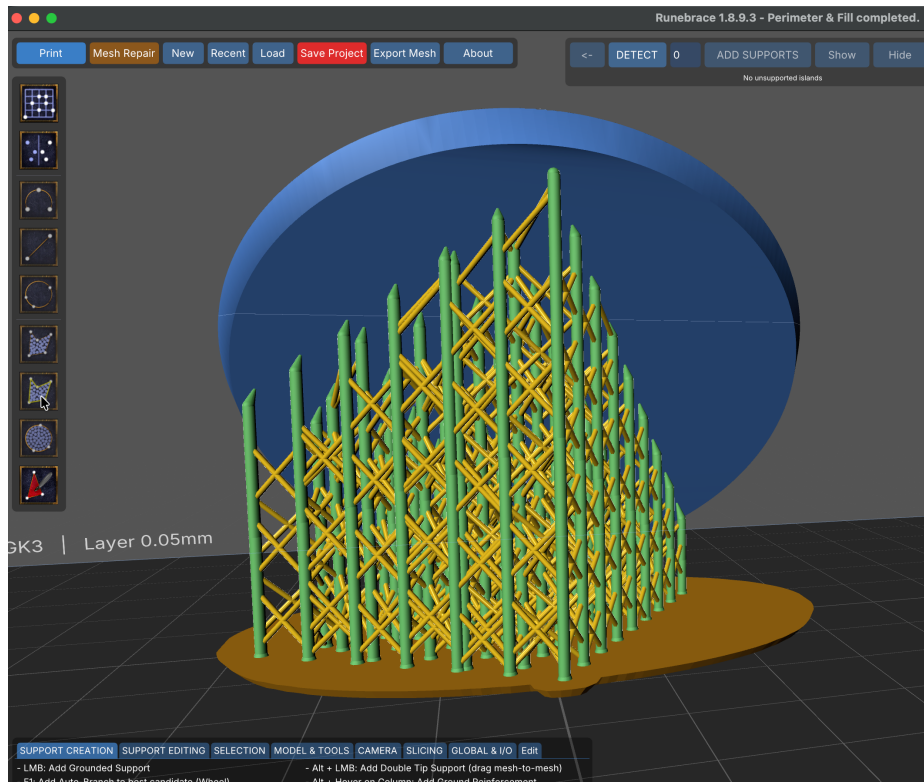


Figure 11: Poly fill with Perimeter

Perimeter and Fill has two separate adjustments:

- **Line:** Spacing between supports on the lines of the outer perimeter
- **Fill Space:** Spacing between supports in the grid that fills the area

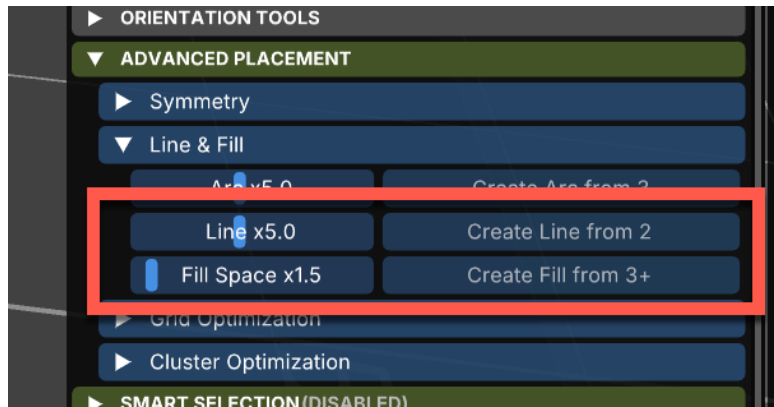


Figure 12: Menu for adjusting both Polygon Fill and Line spacing

These settings are demonstrated in the example below. The perimeter supports are denoted by the red hash lines and were configured with a spacing of 5.0. The inside area was configured with a spacing of 1.5.

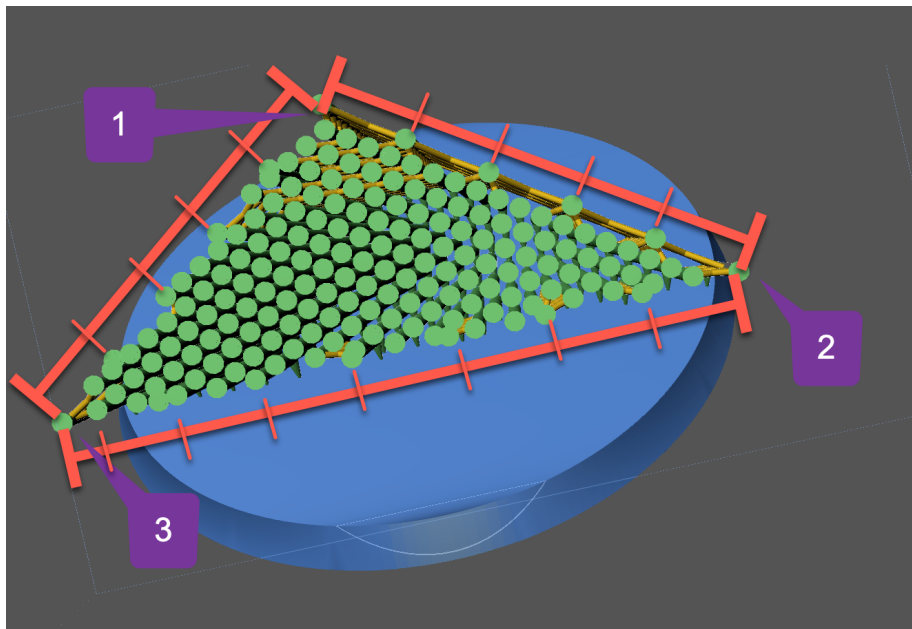


Figure 13: Poly fill with perimeter as seen from underneath

Circle Fill

Circle Fill takes exactly three supports, builds a circular path that passes through their contact points, and fills that. No path supports are created.

Circle Fill is available **only** through the corresponding button on the left button menu (fig. 1).

Using Circle Fill from our starting point results in this:

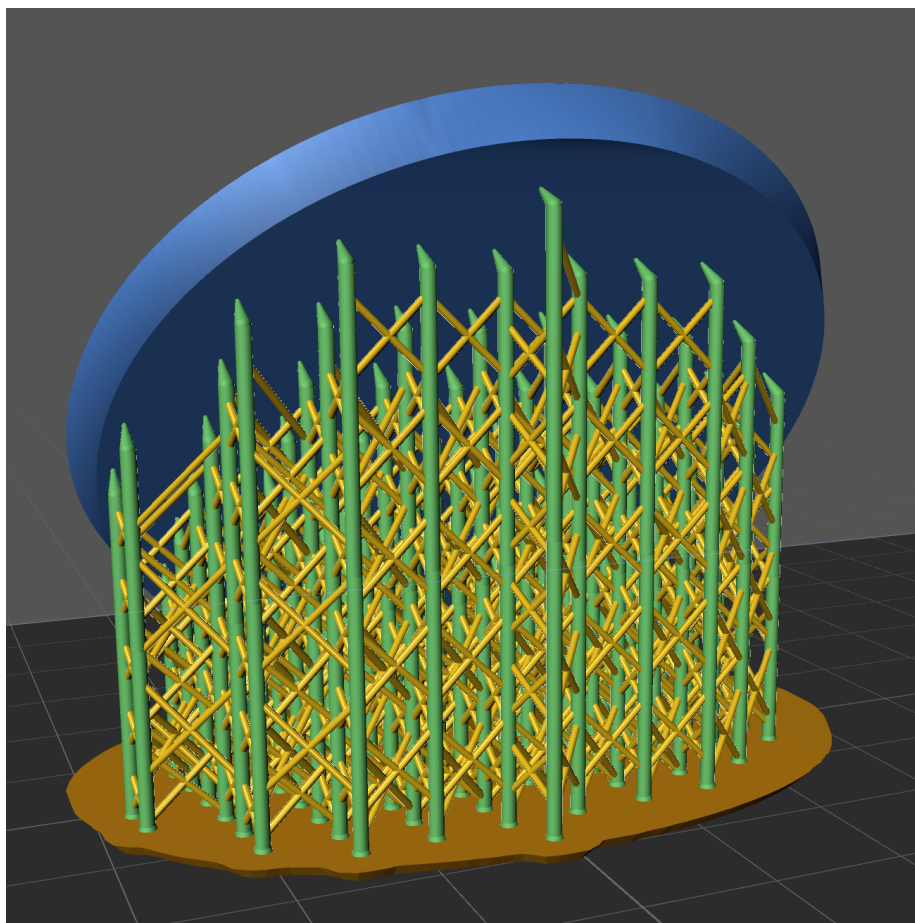


Figure 14: Circle fill

From below, the result looks like:

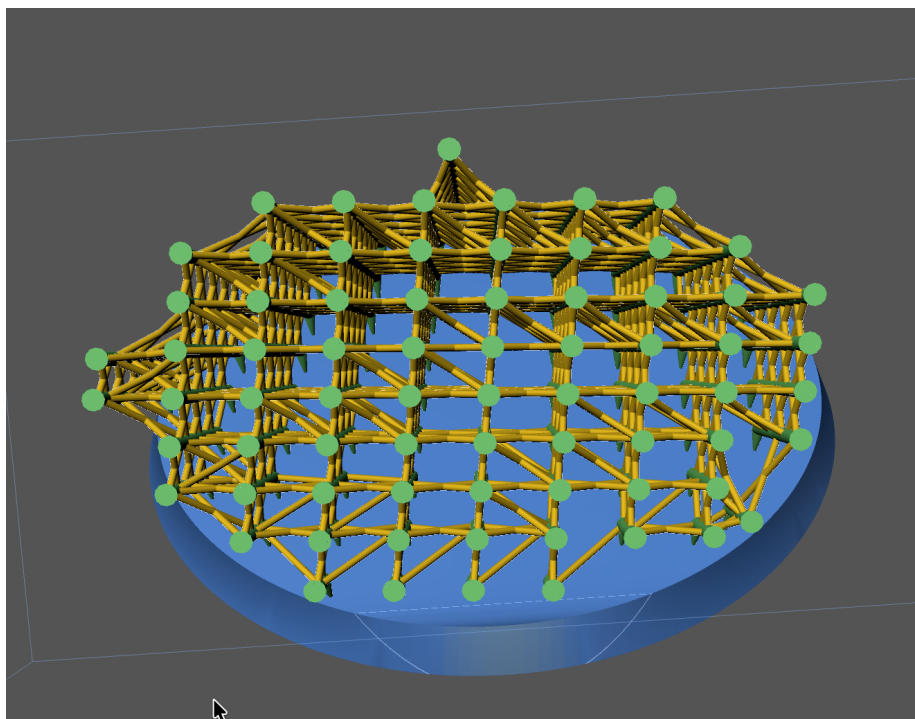


Figure 15: Circle fill viewed from below

Triangle Painting

Rim

Deeper Dive on Placement

lines “What happens when you draw support lines or arcs is that it simulates a mouse clicking on various points along this line (depending on the set distance) and performs a mouse raycast on the model. If the raycast hits the geometry and the face is not parallel to the build plate or tilted upwards, a support is placed. It is likely that there are some raycast misses, so the support cannot be placed and is skipped. It’s not a matter of whether the surface is flat... the geometry must be under the ray cast by the mouse and be a valid point for a support.”

details

” Five details worth having in the docs. The perimeter walks the supports in selection order and closes the loop back to the first one, so the last segment is support N to support 1. The fill plane is defined by the first three supports only, not by all of them: any support beyond the third is flattened onto that

plane, so a selection that is not roughly planar will not fill the way it looks. The grid covers the bounding box of the polygon in that plane and keeps the points that fall inside it, with one axis along support 1 to support 2 and the other perpendicular to it. Each grid point is then projected onto the model along the average tip direction of the selected supports, so the fill follows the surface, and where that ray misses the mesh no support appears, which is why gaps show up over holes or steep overhangs. Finally, Line and Fill Space are multipliers, not distances, and they multiply different reference diameters: Line multiplies the Base diameter, Fill Space multiplies the Top diameter. The same number on both sliders does not give the same spacing on the outline and on the fill. One more practical note: the operation records two undo steps, the perimeter and the fill, so undoing it takes two presses.”

“very small values stop having an effect below a certain point, and a segment too short to fit anything simply gets no supports in between. Happy to give you the exact arithmetic if you want a footnote.”

Math on Determining Placement “step = Base diameter $\times$ Line (or Top diameter $\times$ Fill Space). Gaps = segment length $\div$ step, rounded down. Supports added = gaps $- 1$, spread evenly, so the real spacing is segment length $\div$ gaps. Because the count rounds down, the spacing is always at least the step and can reach almost double it. Yours: $26.73 \text{ mm} \div 4 \text{ gaps} = 6.68 \text{ mm}$, the 6.64 you measured.”

More Complex Examples

You should now understand the basics of creating supports with the various tools. The example above is very simplistic, so let’s use more complex

Presets

Working with Meshes

Hollowing

Printing and Slicing

Glossary

Here are common terms and phrases used in Runebrace.

Path Path applies to Line, Arc, and Circle. For two selected supports, Runebrace builds a path using a raycast between two points. For three selected supports, Runebrace will build a path by creating a curve that passes through all three supports’ contact points.

Rim In triangle mode, “Rim” denotes the border of a painted triangle selection. It is determined from the mesh edges, and can have both an outer border *plus* the borders of any holes inside that particular selection. See also: “Outer Rim Only” and “All boundaries” options in the Triangle Paint panel.